

Minimizing the action: Euler-Lagrange equations deduction by Hamilton variational principle.

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Abstract

The action minimization process to obtain the Euler-Lagrange equations is in fact one of the most important demonstrations of classical physics, this process not only lead us to get a tool that simplifies a lot of mechanical problems but makes explicit the variational nature of physical phenomena and stand a basis to relate energy and action. In this note we will recreate the process as shown in [2] and [3], we also are going to solve the problem of a particle in a potential $V(x) = \frac{1}{2}kx^2$ (The harmonic oscillator) by the analytical way (by using our physical intuition and by solving the ODE) and we are going to solve the problem numerically with an C++ implementation. The numerical implementation and the source of this work is available in our [GitHub Repository](#)

Keywords: Euler-Lagrange, Action, Mechanics, Numerical implementation, Variational

1. INTRODUCTION

The Euler-Lagrange equations are one of the most important family of differential equations on physics and mathematics, it's formulation represents a generalized method for solving variational optimization problems. It was first derived by Leonard Euler on his treatise *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes* by using geometrical considerations[1]. One year after Lagrange wrote a letter to Euler presenting a fully analytic method to reach the same equations[4].

The Euler-Lagrange equations are also a great tool to reach the movement equations of a dynamical problem, this is because on the Euler-Lagrange method you only have to define the lagrangian functional $\mathcal{L} = T + V$, in most problems it reduces to find or propose a potential, when you operate the lagrangian on the E-L equations you reach the movement equations of the problem. This method is considerably simpler than identifying forces and make the sumatories.

In the next sections we are going to show the derivation process, the harmonic oscillator example, the analytical solution of this example, a numerical implementation and we are going to do some conclusions about this formalism.

2. THE ACTION MINIMIZATION PROCESS

The action in physics is defined in the next way [2]:

$$S = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t) dt \quad (1)$$

Where S is called the action $q(t)$, $\dot{q}(t)$ are the generalized coordinates and generalized velocities and t is the time. So a variation of the action will be:

$$\delta S = \delta \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t) dt \quad (2)$$

Or

$$\delta S = \int_{t_1}^{t_2} \mathcal{L}(q + \delta q, \dot{q} + \delta \dot{q}, t) dt \quad (3)$$

For the next step is necessary to do some assumptions, one of them is that at the boundaries our system satisfy $q(t_1) = q_1$ & $q(t_2) = q_2$, we also had to define $q(t, \alpha)$ as:

$$q_1(t, \alpha) = q(t, 0) + \alpha \eta_1(t) \quad (4)$$

$$q_2(t, \alpha) = q(t, 0) + \alpha \eta_2(t)$$

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Which also can be written as:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \delta \dot{q} \right) dt \quad (5)$$

3. THE HARMONIC OSCILATOR

3.1. The analytical way

3.2. The numerical way

4. CONCLUSIONS

5.

REFERENCES

- [1] L. Euler, "Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes, sive solutio problematis isoperimetrici latissimo sensu accepti", *Lausanne & Geneva: Marcum-Michaelem Bousque*, 1744. [Online]. Available: <https://scholarlycommons.pacific.edu/euler-works/65/>.
- [2] L. D. Landau and E. Lifshitz, *Mechanics*. Elsevier, Jan. 1982.
- [3] H. Goldstein, C. P. Poole, and J. L. Safko, *Classical Mechanics*. Addison-Wesley Longman, Jan. 2002.
- [4] *Euler's Correspondence with Joseph Louis de Lagrange*. [Online]. Available: <http://eulerarchive.maa.org/correspondence/correspondents/Lagrange.html>.

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